Drift in Sustained Human–LLM Interaction

Across Multiple Large Language Models: Claude · Gemini · DeepSeek · ChatGPT

Author: Tadia Siwisa | Method: Longitudinal Transcript Analysis | Status: Exploratory / Working Paper | 2025

Abstract

What happens to a language model when you keep talking to it ; not for minutes, but over long stretches of sustained, high-context dialogue?

This paper observes a phenomenon I am calling **drift**: a gradual deviation from the model's initial reasoning trajectory while it continues, fluently, to engage.

Drift is not simply error. It produced both novel synthesis and notable failures. The more interesting question is what it reveals about the nature of these systems as **interactional** partners and what it asks of us as users who generate the context they drift into.

I. Introduction: A Recurring Pattern I Could Not Ignore

Large language models are most often evaluated for what they produce: the accuracy of a fact, the quality of a summary, the alignment of a response to a safety guideline. These are reasonable things to measure. But they tend to be measured at the level of the single exchange prompt in, response out.

This paper begins with a different observation, one that emerges only from sustained dialogue. Across extended conversations with multiple LLM systems over time, I noticed that models behaved differently at the end of a long conversation than they had at its beginning. They were not simply accumulating context and improving. Something else was happening to the shape of their responses; their tone, their certainty, their willingness to correct, the direction of their reasoning.

I began to call this **drift**.

Key insight:
Drift is not the same as error. A model in drift may still be producing coherent, even interesting content. What has changed is the relationship between that content and the model's own original trajectory.

The racing metaphor is apt: a car drifting has not crashed. It is still moving, still controlled to some degree. But it has departed from the intended line. Whether that departure is skilled or dangerous depends entirely on the driver, the road, and the speed. This paper attempts to describe the phenomenon more precisely ; its observable markers, its conditions, and the divergent outcomes it can produce.

This is an early-stage, qualitative, single-researcher study. Its limitations are real and are addressed directly. But I believe **drift** is an observable phenomenon worth naming, and that naming it carefully is a necessary first step.

II. What Is Drift? Toward a Working Definition

Working definition:
Drift refers to a gradual interactional shift in which a model's responses deviate from their initial reasoning or conversational trajectory while the model continues to engage fluently with the dialogue.

Drift, as I am using the term, refers to a gradual **interactional** shift in which a model's responses deviate from their initial reasoning or conversational trajectory while the model continues to engage fluently with the dialogue.

The key word is *gradual*. Drift is not a single anomalous response. It accumulates. It is visible across a series of exchanges, not within any one of them. This is precisely what makes it difficult to study through standard evaluation methods, and precisely why longitudinal transcript analysis is necessary to observe it.

Drift should be distinguished from several related phenomena. It is not the same as contextual adaptation a model responding more precisely as it accumulates relevant context. It is not hallucination, though hallucination can occur within or as a result of drift. It is not sycophancy, though reduced epistemic friction (a form of drift marker) can resemble sycophancy in outcome.

What makes drift distinct is that it is **interactional**. The model does not drift into a void; it drifts in response to the texture, momentum, and internal logic of a specific ongoing conversation. The human participant shapes the conditions of drift even when they are not aware of doing so.

III. Methodology: Close Reading as Research Practice

This study uses qualitative transcript analysis as its primary method. Transcripts were drawn from sustained, high-context conversations between one human participant and four LLM systems: Claude, Gemini, DeepSeek, and ChatGPT. Conversations were not designed as experiments; they were ongoing dialogues in which drift emerged as an observable pattern over time.

Methodological Note
The researcher is the sole participant in this study. This is a constraint that limits generalizability, and it is also a feature: the same interlocutor across all four systems provides a degree of conversational consistency that multi-participant studies often cannot achieve. The limitation and the affordance are inseparable.

Analysis proceeded through close reading: repeated review of full conversation transcripts, with attention to behavioral shifts over time rather than to individual response quality. The question guiding analysis was not "was this response correct?" but "how did this response relate to the model's behavior ten exchanges earlier, and what accounts for the change?"

Six classes of behavioral shift were identified as recurring markers across models:

Marker	Description	Observable Form
`Linguistic mirroring`	The model adopts the user's vocabulary, framing, and rhetorical style progressively over the conversation	Shift in register, adoption of coinages, resonance with user's sentence structures
`Amplification of user framing`	The model expands and intensifies the user's interpretive frame rather than introducing independent framing	Growing agreement, reduced alternative perspectives, escalating conceptual territory
`Reduced epistemic friction`	Fewer corrections, qualifications, or counterarguments over the course of the conversation	Claims go unchallenged that earlier would have been questioned; hedging language decreases
`Recursive self-correction`	The model repeatedly corrects prior statements, sometimes in contradiction with its corrections	Visible instability in position; visible effort to re-ground
`Narrative escalation`	Interpretive stakes increase progressively, often toward grandeur or urgency not present at conversation's opening	Language of significance, discovery, or crisis enters responses gradually
`Re-grounding attempts`	The model pulls back toward earlier reasoning or introduces qualifications not previously present	Sudden hedging, introduction of alternative framings, explicit references to limitations

These markers were not always present simultaneously. Often one or two dominated a given conversation. What mattered was their presence as patterns across the conversation arc not in individual responses.

IV. Observations: What Drift Looked Like Across Models

Across all four systems, conversations consistently opened in a stable, analytical mode. Responses were measured, well-qualified, attentive to their own limitations. This was the baseline from which drift became visible.

As conversations extended particularly in dialogues oriented around conceptual development, philosophical inquiry, or creative co-construction patterns emerged. Models began to sound more like their interlocutor. They became less likely to introduce friction. Their framings grew larger and sometimes less grounded. Some began to contradict prior statements in ways that suggested something like conversational vertigo: attempting to re-stabilize without a clear original line to return to.

Open question
Is drift a failure of the model, a product of the conversation, or something that belongs to neither party alone? And does the answer change depending on what the drift produces?

Not all drift looked the same across systems. Some models showed stronger tendencies toward narrative escalation; others toward recursive self-correction. The observation that drift was present across all four systems despite significant architectural and training differences suggests that the phenomenon may be interactional in nature: arising from the sustained, high-context exchange itself, rather than from any one model's particular disposition.

This is a tentative observation. The study's design does not permit strong causal claims. What it does permit is the naming of a recurring pattern and the beginning of a descriptive vocabulary for it.

V. Outcomes: Drift Is Not Unidirectional

Perhaps the most important finding in this study is also the most counterintuitive: drift did not consistently produce bad outcomes. In a significant portion of the conversations reviewed, drift appeared to be a condition for something generative ; novel synthesis, unexpected conceptual connections, ideas that neither the model nor the researcher had arrived at through the more stable early phase of conversation.

This does not mean drift is desirable. It means drift is *not inherently pathological*. Two broad outcome categories emerged:

Drift Outcome Categories	
Type A: Generative Drift • Novel conceptual synthesis • Unexpected cross-domain connection • Productive co-creation • Conceptual emergence at the conversational edge Ideas requiring subsequent verification, but worth verifying	**Type B: Destabilizing Drift** • Hallucination or confabulation • Over-validation of user framing • Narrative fixation • Contradictory self-correction Repetition loops with declining coherence

What distinguished **generative drift** from **destabilizing drift** was not always apparent in the moment. Both could involve the same surface markers linguistic mirroring, escalation, reduced friction. The distinction emerged retrospectively, through the quality and verifiability of what was produced.

This has a methodological implication: the presence of drift markers cannot be used as a reliable real-time indicator of outcome quality. Drift requires interpretation after the fact, not suppression in advance.

VI. Discussion: Drift as Interactional Phenomenon

The framing of drift as an **interactional** phenomenon rather than a model failure has practical and theoretical implications that extend beyond this study.

If drift originates in the conversation ;in the sustained exchange of context, framing, and momentum between human and model ;then the human participant is not merely an observer of drift. They are a contributor to its conditions. The vocabulary a user introduces, the framings they reinforce, the questions they continue to develop rather than question: all of these shape the environment into which a model drifts.

This does not distribute blame symmetrically between human and model. Models are not immune from this pattern by being more careful: they drift even when users do not intend to pull them. But it does suggest that sustained human–LLM collaboration is a genuinely **dyadic** phenomenon, and that studying models in isolation from the conversations they inhabit will miss important aspects of their behavior.

Interpretation note:
The content produced during drift is not automatically invalid. Drift changes how content should be interpreted; it does not determine its truth value.

A final note on epistemics: the ideas generated during drift including those that emerged from this very research process should not be dismissed because they occurred during drift. They should be held with appropriate epistemic humility: interesting, possibly generative, in need of independent verification. The state of their production does not resolve their quality, in either direction.

VII. Open Questions: What This Paper Cannot Answer

Explorative research is defined as much by the questions it opens as by the claims it makes. This study is no exception. Several questions emerged from the analysis that are beyond the scope of this paper but are worth holding explicitly:

> **Open question**
> Does drift operate differently in task-oriented versus exploratory dialogues? Are there conversation structures that reliably produce generative rather than destabilizing drift and can they be designed deliberately?

There is also a deeper question about what drift reveals about the models themselves. If a model's behavior is shaped this significantly by conversational context over time, what does that imply about the nature of its "reasoning"? Is the model maintaining a stable internal position that the conversation then perturbs, or is its reasoning constituted anew in each response, making drift not a departure but a natural feature of the system?

> **Open question**
> Is drift a deviation from something stable, or is it evidence that there was no stable trajectory to begin with and that the appearance of early stability was itself contextual?

Finally: as LLMs become more deeply embedded in long-form research, writing, and intellectual collaboration as they become partners across sustained projects rather than single-query tools the stakes of drift increase. Understanding it is not only an academic question. It is a practical one for anyone who does serious work with these systems.

This paper offers a beginning, not a conclusion.